

Peripheral Smear — RBC, WBC & Marrow Morphology

RED CELL WING 45%

ANISOCYTOSIS = SIZE (↑RDW) · POIKILOCYTOSIS = SHAPE · POLYCHROMASIA = RETICS

MACRO-OVALOCYTE
B₁₂/FOLATE · MDS

SPHEROCYTE
HS / AIHA

HYPOCHROMIA
IDA · THAL · SIDEROBLASTIC

SCHISTOCYTE
TTP/HUS · DIC · VALVE

SICKLE
HbS

TARGET
LIVER · THAL

POLYCHROMASIA
RETICULOCYTE

STIPPLING
LEAD · THAL

HOOWELL-JOLLY
ASPLENIA

BURR
UREMIA

SPUR CELL
SEVERE LIVER

TEARDROP
MANDATORY MARROW

ELLIPTOCYTE
HE · IRON DEF

NUCLEATED RBC
ASPLENIA · STRESS

ROULEAUX
PARAPROTEIN

AGGLUTINATION
PARAPROTEIN · AIHA

RED CELL WING · 45%

WHITE CELL WING 25%

W1
HYPERSEGMENTED
MEGALOBlastic
HIGH YIELD

W2
VACUOLATED
SEPSIS

W3
DÖHLE
INFECTION

W4
TOXIC GRAN
INFLAMMATION

W5
LARGE GRAN
CHEDIAK-HIGASHI

W6
BAND
LEFT SHIFT

W7
REACTIVE
VIRAL

W8
SMUDGE
CLL
HIGH YIELD

W9
EOSINOPHIL ↑

W10
BASOPHIL ↑
CML

WHITE CELL WING · 25%

BONE MARROW VAULT 30%

BIOPSY
FIRST
H&E
ARCHITECTURE · CELLULARITY · FIBROSIS

ASPIRATE
SECOND
WRIGHT-GIEMSA
MORPHOLOGY · M:E
IRON · FLOW

BLASTS
PANCYTOPENIA
RETICS <0.1%
NUCLEATED RBCs
TEARDROPS + nRBC
STAGING
UNEXPLAINED ANEMIA

CELLULARITY ~50%
FAT% = AGE
M:E 2.5:1
↓ 1:1 = ERYTHROID HYPERPLASIA
↑ >3:1 = MYELOID HYPERPLASIA

H&E
ARCHITECTURE · CELLULARITY · FIBROSIS
BIOPSY BEFORE ASPIRATION

BONE MARROW VAULT · 30%

1. Macro-ovalocyte

WHAT YOU SEE

A framed specimen card of a large egg-shaped (oval) red cell in the Red Cell Wing — the oversized oval shape is the visual tag for megaloblastic anemia (B12/folate, MDS).

WHAT IT ENCODES

Oval macrocytes (large, egg-shaped RBCs) signal MEGALOBLASTIC anemia — B12 or folate deficiency — and MDS. They appear with hypersegmented neutrophils because impaired DNA synthesis affects all marrow lineages. Both deficiencies raise homocysteine; only B12 deficiency also raises methylmalonic acid (MMA).

BOARD TRAP

WRONG answer: treating presumed megaloblastic anemia with folate alone. Discriminator: if it's actually B12 deficiency, folate masks the anemia while subacute combined degeneration of the spinal cord progresses — confirm B12 with MMA and replace B12 first.

2. Spherocyte

WHAT YOU SEE

A framed card of a small, dense, perfectly round red cell with NO central pale spot — losing the bullseye of pallor is what marks it a spherocyte (hereditary spherocytosis or warm AIHA).

WHAT IT ENCODES

Spherocytes are small, round, hyperchromatic RBCs lacking central pallor, from loss of membrane surface area. Two big causes: hereditary spherocytosis (membrane cytoskeleton defect, e.g., ankyrin/spectrin) and warm autoimmune hemolytic anemia (macrophages bite off antibody-coated membrane). MCHC is often high; osmotic fragility/EMA binding test for HS.

BOARD TRAP

WRONG answer: assuming spherocytes always mean hereditary spherocytosis. Discriminator: the direct Coombs (DAT) separates them — POSITIVE = warm AIHA; NEGATIVE with a family history/splenomegaly = hereditary spherocytosis (confirm with EMA binding or osmotic fragility).

3. Schistocyte

WHAT YOU SEE

A framed card of jagged helmet/triangle red-cell fragments — the torn, shredded shape says the cells were physically sheared (microangiopathy: TTP/HUS/DIC, mechanical valve).

WHAT IT ENCODES

Schistocytes (helmet/fragmented cells) mean RBCs are sheared mechanically: microangiopathic hemolytic anemia (TTP, HUS, DIC, HELLP, malignant hypertension, scleroderma renal crisis) or macroangiopathic shear from a prosthetic heart valve. TTP pentad: MAHA + thrombocytopenia + neuro changes + renal + fever (ADAMTS13 deficiency).

BOARD TRAP

WRONG answer: giving a platelet transfusion to a TTP patient with low platelets. Discriminator: in TTP, platelet transfusion 'fuels the fire' (more microthrombi) and is contraindicated unless life-threatening bleeding — treat with urgent plasma exchange. DIC (unlike TTP) has prolonged PT/PTT, low fibrinogen, high D-dimer.

4. Sickle cell

WHAT YOU SEE

A framed card of a stiff crescent/banana-shaped red cell — the sickle shape itself is HbS polymerized on deoxygenation (sickle cell disease).

WHAT IT ENCODES

Sickle/crescent-shaped cells result from HbS polymerization on deoxygenation (Glu→Val substitution at β6). Seen in sickle cell disease, with vaso-occlusive crises, and often accompanied by Howell-Jolly bodies and target cells from autosplenectomy. Confirm with Hb electrophoresis (HbS predominant, no/low HbA in SS disease).

BOARD TRAP

WRONG answer: ordering a sickle solubility test (Sickledex) alone to diagnose disease vs trait. Discriminator: the solubility test is positive in both trait and disease and cannot distinguish them — use Hb electrophoresis to quantify HbS vs HbA and identify trait (HbAS) vs disease (HbSS).

5. Target cell

WHAT YOU SEE

A framed card of a red cell with a central bullseye dot inside the pale ring (a 'target') — extra membrane piled in the middle marks liver disease, thalassemia, or HbC.

WHAT IT ENCODES

Target cells (codocytes) have a central bullseye from excess membrane relative to hemoglobin. Causes — mnemonic 'HALT': Hemoglobinopathies (thalassemia, HbC), Asplenia, Liver disease, Thalassemia. Also seen post-splenectomy.

BOARD TRAP

WRONG answer: interpreting a few target cells as diagnostic of a specific disease. Discriminator: target cells are nonspecific — correlate with MCV (low → thalassemia/HbC) vs liver disease (normal/high MCV, abnormal LFTs) and the rest of the smear/electrophoresis.

6. Basophilic stippling

WHAT YOU SEE

A framed card of a red cell peppered with fine blue dots throughout — the scattered blue ribosomal speckles are the tag for lead poisoning or thalassemia.

WHAT IT ENCODES

Coarse basophilic stippling = punctate blue ribosomal/RNA aggregates throughout the RBC. Classic for LEAD poisoning (lead inhibits rRNA degradation via ferrochelatase/pyrimidine-5'-nucleotidase) and also thalassemia and sideroblastic anemia. Lead poisoning presents with abdominal pain, neuropathy/encephalopathy, microcytic anemia.

BOARD TRAP

WRONG answer: confusing basophilic stippling (diffuse, RNA, lead/thal) with a single Howell-Jolly body (one round DNA remnant, asplenia) or Pappenheimer bodies (iron, sideroblastic). Discriminator: stippling = MULTIPLE fine blue dots throughout the cell; Howell-Jolly = ONE solid purple inclusion = asplenia.

7. Howell-Jolly

WHAT YOU SEE

A framed card of a red cell carrying a single solid round purple dot — one leftover DNA nuclear remnant that a working spleen would have removed, so it signals asplenia.

WHAT IT ENCODES

Howell-Jolly bodies are single, round, dense nuclear (DNA) remnants normally pitted out by a functioning spleen. Their presence indicates ASPLENIA or functional hyposplenism (post-splenectomy, sickle cell autosplenectomy, celiac disease).

BOARD TRAP

WRONG answer: ignoring the infection risk implied by Howell-Jolly bodies. Discriminator: asplenia means high risk of overwhelming sepsis from encapsulated organisms (S. pneumoniae, H. influenzae, N. meningitidis) — these patients need vaccination and prompt antibiotics for febrile illness.

8. Teardrop cell

WHAT YOU SEE

A framed card of a red cell pulled into a teardrop/pear shape — the cell was squeezed exiting a scarred fibrotic marrow, the hallmark of myelofibrosis.

WHAT IT ENCODES

Teardrop cells (dacrocytes) form as RBCs are squeezed exiting a fibrotic/infiltrated marrow. Hallmark of primary MYELOFIBROSIS and other myelophthisic processes (marrow metastases, leukemia), typically with a leukoerythroblastic smear (immature myeloid cells + nucleated RBCs).

BOARD TRAP

WRONG answer: attempting a marrow ASPIRATE first in suspected myelofibrosis. Discriminator: aspiration gives a 'dry tap' because of fibrosis — you need a core BIOPSY (with reticulin stain) to show fibrosis. Primary myelofibrosis is associated with JAK2 mutation.

9. Rouleaux

WHAT YOU SEE

A framed card of red cells stacked neatly like a roll of coins — the coin-stack is caused by sticky paraprotein in the plasma, pointing to multiple myeloma.

WHAT IT ENCODES

Rouleaux = RBCs stacked like coins, caused by high plasma paraprotein/fibrinogen reducing surface charge. Classic for multiple myeloma/MGUS/Waldenström, also seen with marked inflammation. Prompts SPEP/UPEP, serum free light chains, and a high total-protein/elevated ESR.

BOARD TRAP

WRONG answer: mistaking rouleaux for autoagglutination. Discriminator: rouleaux is orderly coin-stacking (paraprotein, dispersible on dilution); agglutination is irregular clumping (cold agglutinin/IgM-mediated AIHA, positive Coombs for C3) — different causes and workup.

10. Hypersegmented neutrophil

WHAT YOU SEE

A framed White Cell Wing card of a neutrophil whose nucleus has 5-6+ lobes — the over-segmented nucleus is the earliest visual sign of megaloblastic (B12/folate) anemia.

WHAT IT ENCODES

Hypersegmented neutrophils (≥5–6 nuclear lobes) are an early, sensitive sign of MEGALOBLASTIC anemia from B12/folate deficiency (impaired DNA synthesis). They can appear before macrocytosis or anemia develop.

BOARD TRAP

WRONG answer: requiring a high MCV before considering B12/folate deficiency. Discriminator: hypersegmented neutrophils can precede a high MCV, and coexisting iron deficiency can normalize the MCV — check B12/folate (and MMA) even with a normal MCV if the smear shows hypersegmentation.

11. Toxic granulation / Döhle

WHAT YOU SEE

A framed card of a neutrophil studded with coarse dark granules plus pale-blue Döhle inclusions — these 'battle scars' mark a neutrophil revved up by severe infection/sepsis.

WHAT IT ENCODES

Reactive/'left-shift' neutrophil changes of severe infection/sepsis: toxic granulation (coarse dark primary granules), Döhle bodies (blue cytoplasmic RER inclusions), cytoplasmic vacuoles, and bands/immature forms. Reflect accelerated marrow output under inflammatory stress.

BOARD TRAP

WRONG answer: interpreting a marked neutrophilia with left shift automatically as leukemia (CML). Discriminator: a leukemoid REACTION (infection) has high leukocyte alkaline phosphatase (LAP) and toxic changes, whereas CML has LOW LAP and the BCR-ABL/Philadelphia chromosome with basophilia.

12. Smudge cell

WHAT YOU SEE

A framed card of a burst, smeared lymphocyte flattened into a smudge — the fragile leukemic lymphocytes of CLL rupture on the slide, leaving these basket smears.

WHAT IT ENCODES

Smudge (basket) cells are fragile lymphocytes ruptured during smearing — characteristic of chronic lymphocytic leukemia (CLL), alongside an absolute lymphocytosis of small mature lymphocytes. Diagnose/confirm by flow cytometry (CD5+, CD19/CD20+, CD23+).

BOARD TRAP

WRONG answer: diagnosing CLL or acute leukemia from smudge cells alone on the smear. Discriminator: smudge cells suggest CLL but flow cytometry is required for diagnosis; do not confuse mature lymphocytosis (CLL) with blasts (acute leukemia) — blasts demand urgent marrow evaluation.

13. Marrow: biopsy vs aspirate

WHAT YOU SEE

The Bone Marrow Vault doorway split into a BIOPSY core (shows architecture/cellularity/fibrosis) and an ASPIRATE smear (shows individual cell morphology) — two complementary marrow samples.

WHAT IT ENCODES

Core BIOPSY preserves marrow architecture — assesses cellularity, fibrosis (reticulin stain), and infiltration/metastases. ASPIRATE smears show individual cell morphology, blast percentage, and are used for flow cytometry, cytogenetics, and the M:E ratio. They are complementary; biopsy is done before aspiration in the sketch's workflow.

BOARD TRAP

WRONG answer: relying on aspirate alone when fibrosis or a 'dry tap' is suspected. Discriminator: fibrotic/infiltrated marrow (myelofibrosis, mets) yields a dry tap and needs the core BIOPSY to show fibrosis/architecture — aspirate cannot quantify fibrosis or cellularity.

14. M:E ratio 2.5:1

WHAT YOU SEE

The marrow tally board reading 'M:E 2.5:1' with blasts and staining noted — the myeloid-to-erythroid ratio counter that normally sits around 2-3:1.

WHAT IT ENCODES

Normal marrow myeloid:erythroid (M:E) ratio is ~2–3:1 (~2.5:1). A HIGH ratio = myeloid hyperplasia or erythroid hypoplasia (infection, CML, pure red cell aplasia); a LOW ratio = erythroid hyperplasia (hemolysis, blood loss recovery, B12/folate replacement) or myeloid suppression. Blasts ≥20% define acute leukemia.

BOARD TRAP

WRONG answer: reading a high M:E ratio as 'too many red cell precursors.' Discriminator: a HIGH M:E means relatively MORE myeloid (or fewer erythroid) cells; erythroid hyperplasia LOWERS the ratio. Always interpret M:E together with overall cellularity and the blast percentage.

15. Hypochromia

WHAT YOU SEE

A framed Red Cell Wing card of a pale red cell with a large, washed-out central pallor zone taking up more than a third of the cell — the thin rim of hemoglobin marks hypochromia.

WHAT IT ENCODES

Hypochromic RBCs have an enlarged area of central pallor ($>1/3$ of the cell diameter) because each cell carries too little hemoglobin, so MCH and MCHC fall. Classic for IRON DEFICIENCY (hypochromic + microcytic with high RDW), and also thalassemia, sideroblastic anemia, and lead poisoning. Iron studies (low ferritin, low TSAT, high TIBC) confirm IDA; sideroblastic anemia shows ringed sideroblasts and Pappenheimer bodies.

BOARD TRAP

WRONG answer: equating hypochromia + microcytosis automatically with iron deficiency. Discriminator: sideroblastic anemia can be hypochromic yet have HIGH iron/ferritin (basophilic stippling, ringed sideroblasts, dimorphic smear), and thalassemia is hypochromic with a NORMAL/low RDW and high RBC count — check ferritin and the Mentzer index before treating with iron.

16. Polychromasia

WHAT YOU SEE

A framed card of a larger, bluish-gray red cell among normal pink ones — the diffuse blue tint flags a young reticulocyte freshly released from a responding marrow.

WHAT IT ENCODES

Polychromatophilic cells are young reticulocytes that retain ribosomal RNA, giving a bluish-gray hue and larger size on Wright stain (RNA confirmed with new methylene blue as a reticulocyte). Increased polychromasia = an appropriate marrow response, seen in HEMOLYSIS and acute blood loss, and explains a raised MCV in those settings. It is the smear correlate of a high reticulocyte count.

BOARD TRAP

WRONG answer: reading the larger blue cells as macrocytes of B12/folate deficiency. Discriminator: polychromasia = reticulocytosis (marrow responding, high retic, LDH/indirect bili up, haptoglobin down in hemolysis), whereas megaloblastic macrocytes are oval with hypersegmented neutrophils and LOW retics — check the reticulocyte count to tell a responding marrow from a failing one.

17. Elliptocyte

WHAT YOU SEE

A framed card of a cigar/pencil-shaped elongated red cell — the elliptical, rod-like form is the tag for hereditary elliptocytosis or, when pencil-thin, iron deficiency.

WHAT IT ENCODES

Elliptocytes (cigar/oval cells) arise from a defective horizontal membrane skeleton (spectrin self-association/protein 4.1 in HEREDITARY ELLIPTOCYTOSIS, autosomal dominant). Thin, pencil-shaped elliptocytes are also a clue to IRON DEFICIENCY, and a few are seen in thalassemia, myelofibrosis, and megaloblastic anemia. Hereditary elliptocytosis is usually mild/asymptomatic; severe variants (hereditary pyropoikilocytosis) cause neonatal hemolysis.

BOARD TRAP

WRONG answer: calling any smear with elliptocytes hereditary elliptocytosis. Discriminator: numerous pencil-shaped cells with microcytosis and low ferritin point to IRON DEFICIENCY, not a membrane disorder — correlate with iron studies, family history, and degree of hemolysis before diagnosing hereditary elliptocytosis.